Homework 5 - EP471
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1. Elliptic: Heat Transport

The steady-state heat transport is governed by the 2D Poisson equation:

$$-\frac{\partial^2 T}{\partial x^2} - \frac{\partial^2 T}{\partial y^2} = Q$$

For the interior nodes, we approximate the second spatial derivatives using a 2nd-order accurate 5-point central difference stencil:

$$-\frac{T_{i-1,j} - 2T_{i,j} + T_{i+1,j}}{\Delta x^2} - \frac{T_{i,j-1} - 2T_{i,j} + T_{i,j+1}}{\Delta y^2} = Q_{i,j}$$

By mapping the 2D grid (i, j) to a 1D array index $k = j \cdot N_x + i$, we rearrange the terms to assemble the linear system $A \cdot T = b$:

$$\left(\frac{2}{\Delta x^2} + \frac{2}{\Delta y^2}\right) T_k - \frac{1}{\Delta x^2} T_{k-1} - \frac{1}{\Delta x^2} T_{k+1} - \frac{1}{\Delta y^2} T_{k-N_x} - \frac{1}{\Delta y^2} T_{k+N_x} = Q_k$$

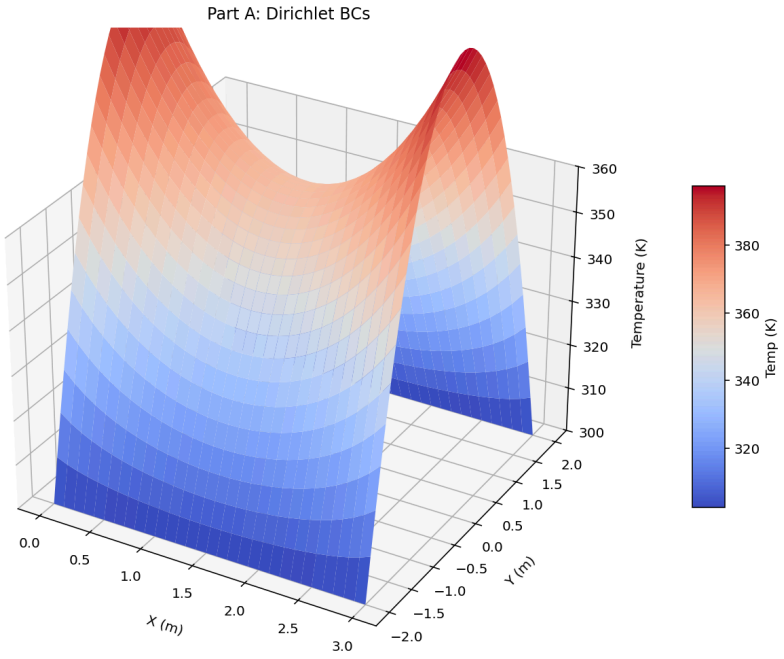
Assumption: The material properties (thermal conductivity) are uniform and normalized to 1, as implied by the given PDE.

Part A:

Dirichlet boundary conditions specify the exact temperature at the boundaries. In our matrix A, the row corresponding to a boundary node k is cleared except for the main diagonal, which is set to 1. The corresponding entry in the vector b is set to the prescribed temperature:

$$A_{k,k} = 1, \quad b_k = T_{boundary}$$

Because these nodes do not couple to their neighbors in the matrix system, the boundary conditions are exactly satisfied.



In a steady-state Poisson system ($-\nabla^2 T = Q$), the final temperature is essentially a superposition of the boundary values and the internal heat generation. The left and right boundaries are defined by $T = 300 + 25(4 - y^2)$. At the centerline of the domain ($y = 0$), the boundaries are held at exactly 400K.

In Figure A, the left and right boundaries are defined by $T = 300 + 25(4 - y^2)$. At the centerline of the domain ($y = 0$), the boundaries are held at exactly 400K.

$T(y) = 300 + 25(4 - y^2)$ is the parabolic distribution of the non-uniform Dirichlet boundaries at x=0 and x=3. As a result, a "thermal ridge" is formed where the domain is pinned to 400K at the center of the side walls (y=0), but decreases to 300K at the corners ($y = \pm 2$). In order to maintain steady-state equilibrium, the source term Q must drive a temperature gradient toward these already hot walls, which is why the internal maximum is higher than the boundaries. The cooling efficiency is low, the domain is being heated from the sides and the middle.

Part B:

Neumann conditions specify the heat flux (temperature derivative) at the boundary. For the top and bottom boundaries, $\frac{\partial T}{\partial y} = 0$.

This can be implemented using a 1st-order accurate finite difference approximation.

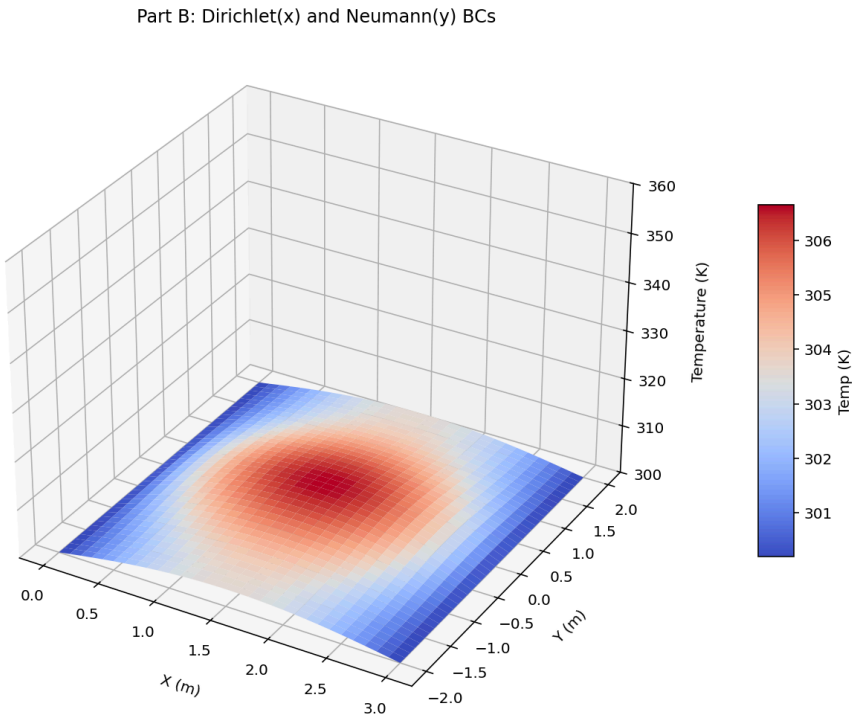
- Bottom Boundary (y = -2): Using a forward difference:

  $$\frac{T_{i,1} - T_{i,0}}{\Delta y} = 0 \implies T_{i,0} - T_{i,1} = 0$$

  In the matrix, this translates to $A_{k,k} = 1$ and $A_{k,k+N_x} = -1$, with $b_k = 0$.
- Top Boundary (y = 2): Using a backward difference:

  $$\frac{T_{i,N_y-1} - T_{i,N_y-2}}{\Delta y} = 0 \implies T_{i,N_y-1} - T_{i,N_y-2} = 0$$

  Matrix entries: $A_{k,k} = 1$ and $A_{k,k-N_x} = -1$, with $b_k = 0$.



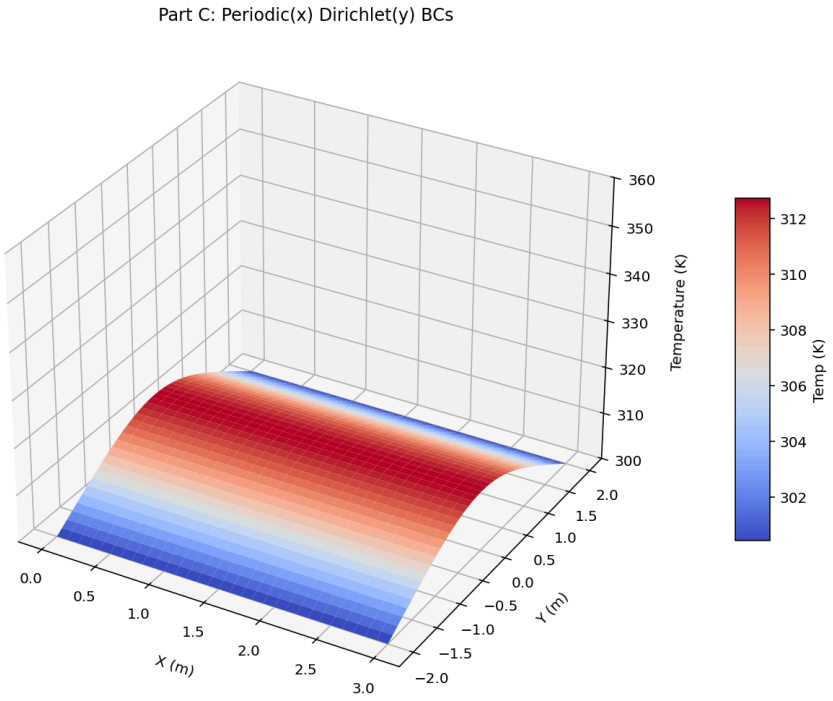
The Neumann BCs were implemented using 1st-order accurate one-sided differences (forward difference for the bottom and backward difference for the top) to maintain the boundary flux at zero without requiring ghost nodes.

Here in Figure B, the Dirichlet boundaries are held at a constant 300K. Using $\frac{\partial T}{\partial y} = 0$ at $y = \pm 2$, we can simulate top and bottom surfaces that are perfectly insulated. Heat can only leave through the left and right boundaries (T=300K), in contrast to Part A, where heat could escape through all four edges. Since a zero gradient indicates that the temperature stays constant across the boundary interface, this can be seen visually in the "flattening" of the surface close to the top and bottom edges.

Part C:

Periodic conditions along the x-direction dictate that the physical state at x=0 is identical to x=3. This is managed conceptually using ghost nodes, where the value on one edge wraps around to the opposite edge. Computationally, since $T_{0,j} = T_{N_x-1,j}$, the node at $i = N_x - 1$ is redundant.

- For the left boundary (i=0), we use the standard 5-point stencil, but the "left neighbor" index is wrapped to the interior node on the right side: $i = N_x - 2$.
- For the right boundary ($i = N_x - 1$), we simply enforce equality with the left boundary: $A_{k,k} = 1$, $A_{k,k-(N_x-1)} = -1$, and $b_k = 0$.



In Figure C, once again, the Dirichlet boundaries are held at a constant 300K, and the heat can easily escape such boundaries. The difference is the ‘wrap around’, the periodic condition along the x-axis effectively simulates an infinitely long plate. Because there are no physical 'walls' at x=0 and x=3 to pin the temperature to a specific value, the solution becomes independent of x and varies only with y. The profile resembles a 1D solution 'extruded' along the x-direction, with the only heat sinks being the Dirichlet boundaries at the top and bottom.

Part D:
In plotting the error for the boundary conditions for part C, we see a slope of around 2. This confirms that the central difference scheme used for the interior nodes is second-order accurate. Since Part D uses the Periodic conditions from Part C (which are essentially interior stencils), the global accuracy should maintain its second-order property.

